19.3: Viruses and prions are formidable pathogens in animals and plants

Diseases caused by viral infections afflict humans, agricultural crops, and livestock worldwide. Other smaller, less complex entities known as prions also cause disease in animals. We'll first consider animal viruses.

Viral Diseases in Animals

A viral infection can produce symptoms by a number of different mechanisms. Viruses may damage or kill cells by causing the release of hydrolytic enzymes from lysosomes. Some viruses cause infected cells to produce toxins that lead to disease symptoms, and some have molecular components that are toxic, such as envelope proteins. How much damage a virus causes depends partly on the ability of the infected tissue to regenerate by cell division. People usually recover completely from colds because the epithelium of the respiratory tract, which the viruses infect, can efficiently repair itself. In contrast, damage inflicted by poliovirus to mature nerve cells is permanent because these cells do not divide and usually cannot be replaced. Many of the temporary symptoms associated with viral infections, such as fever and body aches, actually result from the body's own efforts to defend itself against infection rather than from cell death caused by the virus.

The immune system is a critical part of the body's natural defenses (see Chapter 43). It is also the basis for the major medical tool used to prevent viral infections—vaccines. A **vaccine** is a harmless component of a pathogen that stimulates the immune system to mount defenses against the harmful pathogen. Smallpox, a viral disease that was once a devastating scourge in many parts of the world, was eradicated by 1980 due to a vaccination program carried out by the World Health Organization (WHO). The very narrow host range of the smallpox virus—it infects only humans—was a critical factor in the success of this program. Similar worldwide vaccination campaigns are under way to eradicate polio, the incidence of which has dropped by 99%, and measles. Although an effective vaccine exists for measles, a large outbreak occurred in the Pacific Northwest in 2019, correlated with lower vaccination rates in that region. Effective vaccines are also available to protect against rubella, mumps, hepatitis B, and a number of other viral diseases. Less than a year after SARS-CoV-2 was identified as the cause of COVID-19, multiple vaccines were developed and approved for use. Many nations immediately initiated programs to administer shots rapidly to their populations. Unfortunately, many less-resourced countries lagged behind in receiving needed vaccine doses, one factor contributing to the persistence of the pandemic.

Although vaccines can prevent some viral illnesses, medical care can do little, at present, to cure most viral infections once they occur. The antibiotics that help us recover from bacterial infections are powerless against viruses. Antibiotics kill bacteria by inhibiting enzymes specific to bacteria but have little or no effect on eukaryotic or virally encoded enzymes. However, the few enzymes that are encoded only by viruses have provided targets for other drugs. Most antiviral drugs resemble nucleosides and thus interfere with viral nucleic acid synthesis. One such drug is acyclovir, which impedes herpesvirus replication by inhibiting the viral polymerase that synthesizes viral DNA but not the eukaryotic one. Similarly, azidothymidine (AZT) curbs HIV replication by interfering with the synthesis of DNA by reverse transcriptase. In the past 30 years, much effort has gone into developing drugs to treat HIV. Currently, multidrug treatments, called *antiretroviral therapy* or *ART*, are considered to be most effective. Such treatments commonly include a combination of two nucleoside mimics and a protease inhibitor, which interferes with an enzyme required for assembly of the viruses. Multidrug treatments originally involved taking up to 20 pills multiple times per day but now usually consist of a single daily tablet. Another effective treatment involves a drug called maraviroc, which blocks a protein on the surface of human immune cells that helps bind the HIV virus (see Figure 7.8). This drug has also been used successfully to prevent infection in individuals who either have been exposed to, or are at risk of exposure to, HIV. In a similar approach, the current vaccines against SARS-CoV-2 are designed to raise an immune response against the Spike protein on the viral surface, blocking it from binding to the cell surface receptor during infection.

BBC Video

Know Your Enemy: Bacteria vs. Viruses

image

image

HHMI BioInteractive Video

Treating HIV Infection with a Reverse Transcriptase Inhibitor

image

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Emerging Viral Diseases

Viruses that suddenly become apparent are often referred to as *emerging viruses*. HIV, the AIDS virus, is a classic example: The earliest HIV infections of humans occurred in sub-Saharan Africa, perhaps as early as the 1920s or 1930s, with outbreaks in the 1960s and 1970s. In the United States, the first cases were reported in the early 1980s among young gay men in California and New York. In 1982, the Centers for Disease Control named the disease AIDS. Another example of an emerging virus is West Nile virus, which causes encephalitis, inflammation of the brain. This appeared in North America in 1999 and has spread to 49 U.S. states, by 2019 resulting in over 50,000 cases and about 2,000 deaths.

The deadly Ebola virus **(Figure 19.10a)**, recognized initially in 1976 in central Africa, is one of several emerging viruses that cause *hemorrhagic fever*, an often fatal illness characterized by fever, vomiting, massive bleeding, and circulatory system collapse. In 2014, a widespread outbreak, or **epidemic**, of Ebola virus occurred. By 2016, it had resulted in over 11,000 deaths. In 2017, 2018, and 2019, smaller outbreaks occurred in the Democratic Republic of the Congo.

Figure 19.10

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Emerging viruses.
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Three photos of emerging viruses.

The viruses are as follows.

Part a. Ebola viruses budding from a monkey cell (colorized S E M). Scale 500 nanometers.

Part b. Chikungunya viruses emerging from a cell in the upper left and packing together (colorized T E M). Scale 200 nanometers.

Part c. Computer-generated image of a Zika virus, based on a technique called cryo-electron microscopy. Scale 20 nanometers.

The mosquito-borne chikungunya virus **(Figure 19.10b)** causes an acute illness with fever, rashes, and persistent joint pain. Chikungunya has long been considered a tropical virus, but it has now appeared in Italy, France, and Spain. The Zika virus **(Figure 19.10c)** was first observed in Uganda in 1947, but for decades only a few cases occurred per year. In the spring of 2015, however, it became an emerging virus when it caused a large outbreak in Brazil. Although symptoms of Zika are often mild, the infection of pregnant individuals was correlated with a striking increase in the number of babies born with abnormally small brains, a condition called microcephaly. Zika is a mosquito-borne flavivirus (like West Nile virus) that infects neural cells, posing a particular danger to fetal brain development.

HHMI BioInteractive Video

Interview with Katie Walter

image

image

HHMI BioInteractive Video

Genetically Modified Mosquitoes

image

image

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Where do these new strains of virus come from? One cause of rapidly emerging viral diseases in humans is mutation of existing viruses into new viruses that can spread more easily. RNA viruses have a high rate of mutation because viral RNA polymerases do not proofread and correct errors in replicating their RNA genomes. Some mutations change existing viruses into new viral strains that can cause disease, even in people immune to the original virus. A well-known related example is how three or four mutations causing changes in a surface protein of a cat virus (feline panleukopenia virus) resulted in the emergence in 1978 of canine parvovirus, a very contagious deadly virus infecting dogs.

A second cause of the emergence of viral diseases is the spread of a viral disease from a small, isolated human population. AIDS went unnamed and virtually unnoticed for decades before spreading around the world, accelerated by affordable international travel. HIV is transmitted through exchange of bodily fluids; this happens, for instance, through blood transfusions, unprotected sexual intercourse, and reuse of needles to inject intravenous drugs. In African countries, lack of resources and stigmatization of infected individuals initially hastened the spreading of AIDS. In the United States, AIDS was seen at first as only affecting gay men. Societal prejudice toward this community led to a lack of resources to investigate and stem the spread of the disease, which contributed to its worldwide spread.

Interview

Interview with David Satcher: The role of the CDC in recognizing AIDS and in public health

David Satcher.

A third cause of new viral diseases in humans is the spread of existing viruses from other animals. Scientists estimate that about three-quarters of new human diseases originate in this way. Animals infected with a virus that can be transmitted to humans are said to be a natural reservoir for that virus. HIV is an example,

as scientists believe that it originated from a version of the virus found in chimpanzees in central Africa, after people ate chimpanzee meat for food and were infected by exposure to chimpanzee blood. In a more recent example, many scientists think that bats passed SARS-CoV-2 to humans, possibly via an intermediate host.

In general, flu epidemics provide an instructive example of these three causes of emerging viruses. There are three types of influenza virus: types B and C, which infect only humans and have never caused an epidemic, and type A, which infects a wide range of animals, including birds, pigs, horses, and humans. The influenza type A viruses present in pigs and wild and domestic birds are potential emerging viruses that represent a long-term threat to human health.

A case in point is the H5N1 strain of avian influenza virus, which is highly contagious and deadly in birds. The first transmission from birds to humans was documented in Hong Kong in 1997. Since then, about 700 people have been infected, with an alarming mortality rate of greater than 50%. The high mortality rate is partly because H5N1 is very different from strains of influenza that have circulated among humans for a long time. Individuals are therefore not able to mount a strong immune response against viruses like H5N1.

Different strains of influenza A are given standardized names; for example, the name H5N1 identifies which forms of two viral surface proteins are present—hemagglutinin and neuraminidase (HA and NA, respectively; see the glycoprotein spikes in Figure 19.3c). These two proteins together help determine the host range and severity of disease caused by each virus. As of 2017, 18 types of hemagglutinin, a protein that helps the flu virus attach to host cells, and 11 types of neuraminidase, an enzyme that helps release new virus particles from infected cells, have been identified. All possible combinations of HA and NA have been found in some waterbirds. In 2018, researchers using advanced microscopy techniques reported that the HA protein is quite dynamic, stretching toward the target host cell, retracting, then refolding to a new shape and re-approaching the target cell.

Although deadly, the H5N1 strain has not yet caused an epidemic because nearly all cases have been the result of transmission from birds to people, rather than person-to-person. Epidemics occur when genetic changes allow a new viral strain to be easily transmitted between humans. An event like this occurred in 2009, when a strain of influenza virus (H1N1) appeared that was very different from the virus that causes the seasonal flu. The H1N1 influenza virus spread rapidly, prompting WHO to declare a global epidemic, or **pandemic**. Within half a year, the disease had reached 207 countries, infecting over 600,000 people and killing almost 8,000. In a more recent example, the first case of COVID-19 was reported in Wuhan, China, in late December 2019, and on March 11, 2020, WHO declared it a pandemic. Within the next 18 months, virtually every country was battling SARS-CoV-2.

In addition to the 2009 H1N1 pandemic, influenza A strains have caused three other major flu epidemics among humans in the last 100 years. The most notable of these was the “Spanish flu” pandemic of 1918–

1919, which killed 40–50 million people worldwide. In the **Scientific Skills Exercise**, you'll analyze genetic changes in variants of the 2009 H1N1 influenza virus and correlate them with spread of the disease.

Scientific Skills Exercise

Analyzing a Sequence-Based Phylogenetic Tree to Understand Viral Evolution

AP Science Practices 2, 4, 5, 6

How Can Sequence Data Be Used to Track Flu Virus Evolution? In 2009, an influenza A H1N1 virus caused a pandemic, and the virus has continued to resurface in outbreaks across the world. Researchers in Taiwan were curious about why the virus kept appearing despite widespread flu vaccine initiatives. They hypothesized that newly evolved variant strains of the H1N1 virus were able to evade human immune system defenses. To test this hypothesis, they needed to determine if each wave of flu infection was caused by a different H1N1 variant strain.

A nurse injects a patient's arm with a vaccine.

*H1N1**flu vaccination***

How the Experiment Was Done Scientists obtained the genome sequences for 4,703 virus isolates collected from patients with H1N1 flu in Taiwan, each named by type/location/identifying number/year. They compared the sequences in different strains for the viral hemagglutinin (HA) gene, and based on mutations that had occurred, arranged the isolates into a phylogenetic tree (see Figure 26.5 for information on how to read phylogenetic trees).

Data from the Experiment In the phylogenetic tree, each branch tip is one variant strain of the H1N1 virus with a unique HA gene sequence. The tree is a way to visualize a working hypothesis about the evolutionary relationships between H1N1 variants.

A phylogenetic tree.

Parts are as follows. Phylogenetic tree: each branch tip is one variant strain of the H 1 N 1 virus with a unique H A gene sequence. From the left, there are three main branch points (nodes) splitting. Starting at the top of the tree as drawn and moving down through the tree, passing through five nodes from the left and sharing the last node is the following.

A/California/07/2009 (Group 1) and

- A/Taiwan/1164/2010 (Group 3)

Passing through four nodes is:

- A/Taiwan/T1773/2009 (Group 6)

From the third node are (all Group 7, top to bottom as drawn):

- A/Taiwan/T1338/2009
- A/Taiwan/T0724/2009
- A/Taiwan/T1821/2009
- A/Taiwan/937/2009
- A/Taiwan/T1339/2009

Then another node to:

- A/Taiwan/940/2009 and
- A/Taiwan/7418/2009

Then back to the third node for:

- A/Taiwan/8575/2009

Followed by another node branching from the third node to:

- 2 nodes from 3rd node: A/Taiwan/4909/2009 and A/Taiwan/8542/2009

And two nodes branching from third node to (both Group 9):

- A/Taiwan/1018/2011 and
- A/Taiwan/552/2001

Then back to third node for:

- A/Taiwan/2826/2009 (Group 7)

Second node from left branches to:

- A/Taiwan/T0826/2009 (Group 7)

And branching from the first node from the left:

- Through 2 additional nodes: A/Taiwan/1017/2009 (Group 8) and A/Taiwan/7873/2009 (Group 8)

Then, back to branching from the first node are:

- A/Taiwan/11706/2009 (Group 8)
- A/Taiwan/6078/2009 (Group 8)
- A/Taiwan/6341/2009 (Group 8)
- A/Taiwan/6200/2009 (Group 8)

Then an additional node branches to:

- A/Taiwan/5270/2010 (Group 8-1) and
- A/Taiwan/3994/2010 (Group 8-1)

Below, a long branch from the left goes to one node branching to

- A/Taiwan/2649/2011 and A/Taiwan/1102/2011 (both Group 10)

Below which another region of heavy branching starts at the left and goes, through 3 nodes to (all Group 11):

- A/Taiwan/4501/2011 and
- A/Taiwan/67/2011

Then through two nodes from the left is:

- A/Taiwan/1749/2011

Below, from a different branch point but also through 2 nodes from the left is:

- A/Taiwan/4611/2011

Below, through three nodes is:

- A/Taiwan/5506/2011

Through four nodes is:

- A/Taiwan/1150/2011

Through 5 nodes is:

- A/Taiwan/2883/2011

And through a 6th node (sharing this node) are:

- A/Taiwan/842/2010 and
- A/Taiwan/3697/2011

INTERPRET THE DATA

- The more closely connected two variants are in the tree, the more alike they are in terms of HA gene sequence. Each fork in a branch, called a node, shows where two lineages separate due to different accumulated mutations. The length of the branches is a measure of how many sequence differences there are between the variants, indicating how distantly related they are. Referring to the phylogenetic tree, which variants are more closely related to each other: A/Taiwan/1018/2011 and A/Taiwan/552/2011 or A/Taiwan/1018/2011 and A/Taiwan/8542/2009? Explain your answer.
- The scientists arranged the branches into groups made up of one ancestral variant and all of its descendant, mutated variants. They are color-coded in the figure. Using group 11 as an example, trace the lineage of its variants.
-
    - Do all of the nodes have the same number of branches or branch tips?
- Are all of the branches in the group the same length?
- What do these results indicate?
- The graph shows the number of isolates collected (each from an ill patient) on the y-axis and the month and year that the isolates were collected on the x-axis. Each group of variants is plotted separately with a line color that matches the tree diagram.
-
    - Which group of variants was the earliest to cause the first wave of H1N1 flu in over 100 patients in Taiwan?
- After a group of variants had a peak number of infections, did members of that same group cause another (later) wave of infection?
- One variant in group 1 (green, uppermost branch) was used to make a vaccine that was distributed very early in the pandemic. Based on the graphed data, does it look like the vaccine was effective?

A graph of viral variants.

Scientists graphed the number of isolates by the month and year of isolate collection to show the period in which each viral variant was actively causing illness in people.

Data from J.-R. Yang et al., New variants and age shift to high fatality groups contribute to severe successive waves in the 2009 influenza pandemic in Taiwan, *PLoS ONE*6(11): e28288 (2011).

Graph of the number of viral isolates versus months between 2009 and 2011. Y-axis label is number of viral isolates from 0 to 800. 4 waves of the viral isolates are shown.

Wave 1: May to October, 2009 which includes Groups 1, 3, 6 with a peak of about 20 and Group 7 with a peak of about 700 – both peaks occur in August;

Wave 2: October to February 2010 which includes Group 8 with a peak of about 625 in November;

Interwave: February 2010 to November 2010 which includes Group 8-1 with a peak of about 50 in August (all other groups are flat);

Wave 3: December 2010 to April 2011 includes Group 9 (peak about 75), Group 10 (peak about 250), and Group 11 (peak about 500) all of which peak in January.

- Groups 9, 10, and 11 all had H1N1 variants that caused a large number of infections at the same time in Taiwan. Does this mean that the scientists' hypothesis, that new variants cause new waves of infection, was incorrect? Explain your answer.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

The disease caused by H1N1 was originally called "swine flu" because parts of the viral genome were very similar to strains of influenza in pigs. However, studies revealed that the virus was not transmitted from pigs to humans. Instead, the story was more complex: H1N1 was a unique combination of swine, avian, and human influenza genes that allowed it to spread among humans.

Influenza viruses can change quickly because they have a genome made up of eight segments of RNA rather than a single RNA molecule (see Figure 19.3c). When an animal like a pig or a bird is infected with multiple strains of influenza virus, the RNA molecules making up the viral genomes can mix and match (*reassort*) during viral assembly, resulting in new genetic combinations. If a flu virus from pigs recombines with viruses that circulate widely among humans, it may acquire the ability to spread easily from person to person, dramatically increasing the potential for a major human outbreak. Pigs are believed to have been the main hosts for recombination that led to the 2009 H1N1 flu virus.

Influenza viruses also have a high rate of mutation, for reasons mentioned earlier. Coupled with reassortments, these mutations can lead to the emergence of a viral strain from animals that can infect human cells. Scientists are worried that the H5N1 strain will evolve in a way that enables it to spread as easily as the H1N1 strain. This would represent another major global health threat like that of the 1918 influenza pandemic.

How easily could this capability evolve in the H5N1 strain? In 2011, scientists working with ferrets, small mammals used as model organisms for the human flu, found that only a few mutations of the avian flu virus were sufficient to allow infection of cells in the human nasal cavity and windpipe. Furthermore, when the scientists transferred nasal swabs serially from ferret to ferret, the virus became transmissible through the air. Reports of this startling discovery ignited a firestorm of debate about whether to publish the results. Ultimately, the scientific community decided the benefits of preventing pandemics outweighed the risks of the information being used for harmful purposes, and the work was published in 2012.

In fact, we have observed a similar phenomenon in SARS-CoV-2 during the COVID-19 pandemic. Widespread SARS-CoV-2 infection of many individuals has allowed random mutations to occur, and evolutionary processes have led to new, more easily transmissible strains of the virus emerging and becoming predominant. Clearly, the COVID-19 pandemic has sharply highlighted the crucial importance of understanding the evolution and functioning of viruses.

Normal seasonal flu viruses (including influenza types A and B) are not considered emerging viruses because variations of seasonal flu viruses have been circulating among humans for long enough that most components are recognized by the immune system. However, these viruses still undergo mutation and reassortment of genome segments, and variations of the HA protein are used each year to generate vaccines against the strains predicted most likely to occur the following year.

As we have seen, emerging viruses are usually existing viruses that mutate, spread more widely in the current host species, or spread to new host species. Changes in host behavior or environmental changes can increase the viral traffic responsible for emerging diseases. For instance, new roads built through remote areas can allow viruses to spread between previously isolated human populations. Also, the destruction of forests to expand cropland can bring humans into contact with animals that host infectious viruses. Finally, genetic mutations and changes in host ranges can allow viruses to jump between species. Many viruses are transmitted by mosquitoes. A dramatic expansion of the disease caused by the chikungunya virus occurred in the mid-2000s when a mutation allowed it to infect not only the mosquito species *Aedes aegypti* but also the related *Aedes albopictus*. Insecticides and mosquito netting over beds are crucial tools in public health attempts to prevent diseases carried by mosquitoes **(Figure 19.11)**.

Figure 19.11

Netting as protection against virus-carrying mosquitoes.

Photo of baby sleeping inside mosquito netting. Close-up photo of a mosquito biting human skin.

Recently, scientists have become concerned about the possible effects of climate change on worldwide viral transmission. Dengue fever, also mosquito-borne, has appeared in Florida and Portugal, regions where it had not been seen before. The possibility that global climate change has allowed mosquito species carrying these viruses to expand their ranges and interact more is troubling because of the increased chance of a mutation allowing a virus to jump to a new host. This is an area of active research by scientists applying climate change models to what is known about the habitat requirements of mosquito species.

Prions: Proteins as Infectious Agents

The viruses discussed in this chapter are infectious agents that spread diseases, and their genetic material is composed of nucleic acids, whose ability to be replicated is well known. Surprisingly, there are also *proteins* that are infectious. Proteins called **prions** appear to cause degenerative brain diseases in various animal species. These diseases include scrapie in sheep; mad cow disease, which plagued the European beef industry about 20 years ago; and Creutzfeldt-Jakob disease in humans, which has caused the death of some 175 people in the United Kingdom since 1996. Prions can be transmitted in food, as may occur when people eat beef from cattle with mad cow disease. Kuru, another human disease caused by prions, was identified in the early 1900s among the South Fore indigenous people of New Guinea. When a kuru epidemic peaked there in the 1960s, scientists at first thought the disease had a genetic basis because family members also often contracted the disease. Eventually, however, investigations revealed a different story: After a death, family members practiced ritual cannibalism, eating organs of the deceased, and prions were transmitted primarily in brain tissue. Women got kuru more often than men because men ate the more “prestigious” organs, like the heart, while women and children ate the brains.

Two characteristics of prions are especially alarming. First, prions act very slowly, with an incubation period of at least ten years before symptoms develop. The lengthy incubation period prevents sources of infection from being identified until long after the first cases appear, allowing many more infections to occur. Second, prions are not destroyed or deactivated by heating to normal cooking temperatures. To date, there is no known cure for prion diseases, and the only hope for developing effective treatments lies in understanding the process of infection.

How can a protein, which cannot replicate itself, be a transmissible pathogen? According to the leading model, a prion is a misfolded form of a protein normally present in brain cells. When the prion gets into a cell containing the normal form of the protein, the prion somehow converts normal protein molecules to the misfolded prion versions. Several prions then aggregate into a complex that can convert other normal proteins to prions, which join the chain **(Figure 19.13)**. Prion aggregation interferes with normal cellular functions and causes disease symptoms. This model was greeted with much skepticism when it was first proposed by Stanley Prusiner in the early 1980s, but it is now widely accepted. Prusiner was awarded the

Nobel Prize in 1997 for his work on prions. He has also proposed that prions are involved in neurodegenerative diseases such as Alzheimer's and Parkinson's disease. There are many outstanding questions about these small infectious agents.

Figure 19.13

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Model for how prions propagate.
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Prion contacts a normally folded version of the same protein which then assumes the abnormal shape of the prion. This can continue until large aggregates of prions are formed.

Prions are misfolded versions of normal brain proteins. When a prion contacts a normally folded version of the same protein, it may induce the normal protein to assume the abnormal shape. The resulting chain reaction may continue until high levels of prion aggregation cause cellular malfunction and eventual degeneration of the brain.

Animation

Prions: Characteristics

image

Play

image

Animation

Prions: Diseases

image

Play

image

Viral Diseases in Plants

More than 2,000 types of viral diseases of plants are known, accounting for an annual loss of over $30 billion worldwide due to destruction of crops. Common signs of viral infection include bleached or brown spots on leaves and fruits **(Figure 19.12)**, stunted growth, and damaged flowers or roots, all of which can diminish the yield and quality of crops.

Figure 19.12

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Immature tomato infected by a virus.
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Photo of an underdeveloped pale tomato with blotchy brown spots.

Plant viruses have the same basic structure and mode of replication as animal viruses. Most known plant viruses, including tobacco mosaic virus (TMV), have an RNA genome. Many have a helical capsid, like TMV, while others have an icosahedral capsid (see Figure 19.3b).

Viral diseases of plants spread by two major routes. In the first route, *horizontal transmission*, an external source infects the plant. Because the invading virus must get past the plant's outer protective layer of cells (the epidermis), a plant becomes more susceptible to viral infections if it has been damaged by wind, injury, or herbivores. Herbivores, especially insects, pose a double threat because they can also carry viruses, transmitting disease from plant to plant. Moreover, gardeners may transmit plant viruses inadvertently on pruning shears and other tools. The other route of viral infection is *vertical transmission*, in which a plant inherits a viral infection from a parent. Vertical transmission can occur in asexual propagation (for example, through cuttings) or in sexual reproduction via infected seeds.

Once a virus enters a plant cell and begins replicating, viral genomes and associated proteins can spread throughout the plant through plasmodesmata, the cytoplasmic connections that penetrate the walls between adjacent plant cells (see Figure 36.19). The passage of viral macromolecules from cell to cell is facilitated by virally encoded proteins that cause enlargement of plasmodesmata. Scientists have not yet devised cures for most viral plant diseases, so research efforts are focused largely on reducing disease transmission and on breeding resistant varieties of crop plants.